

Honors Intro Micro | Homework 3

Due Thursday Oct. 14

Homework is designed to both test your knowledge and challenge you to apply familiar concepts in new applications. Send me your questions at taylorjweidman@pitt.edu. Answer clearly and completely; show your work so I can understand your thought process for partial credit; you are welcomed and encouraged to work in groups as long as your work is your own.

Part 1

Last homework we looked at the demand curve for pumpkin pasties, which can be represented by:

$$P = 17 - \frac{1}{6}Q$$

Use a graph to plot this demand curve, then find and label the consumer surplus at both 5 galleons and 10 galleons.



CS at 5 galleons:_____

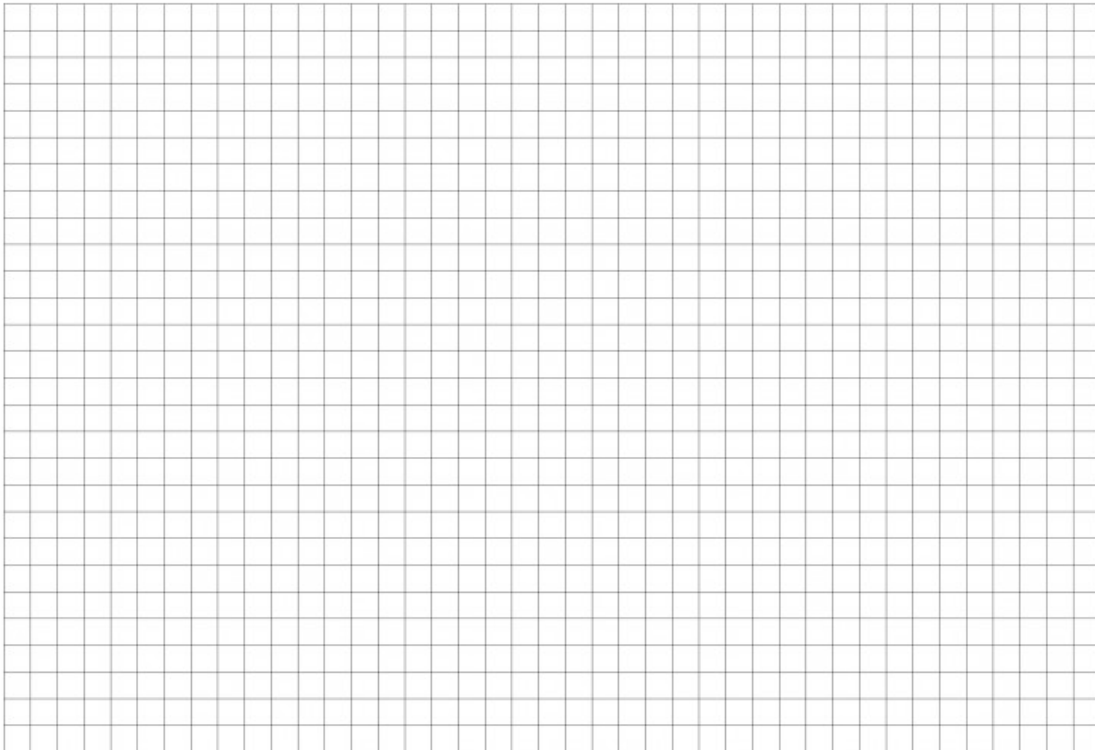
CS at 10 galleons:_____

Part 2

The supply curve for pumpkin pasties can be represented by the equation:

$$P = 2 + \frac{2}{3}Q$$

Use a graph to plot this supply curve, and find and label the producer surplus at both 5 galleons and 10 galleons.

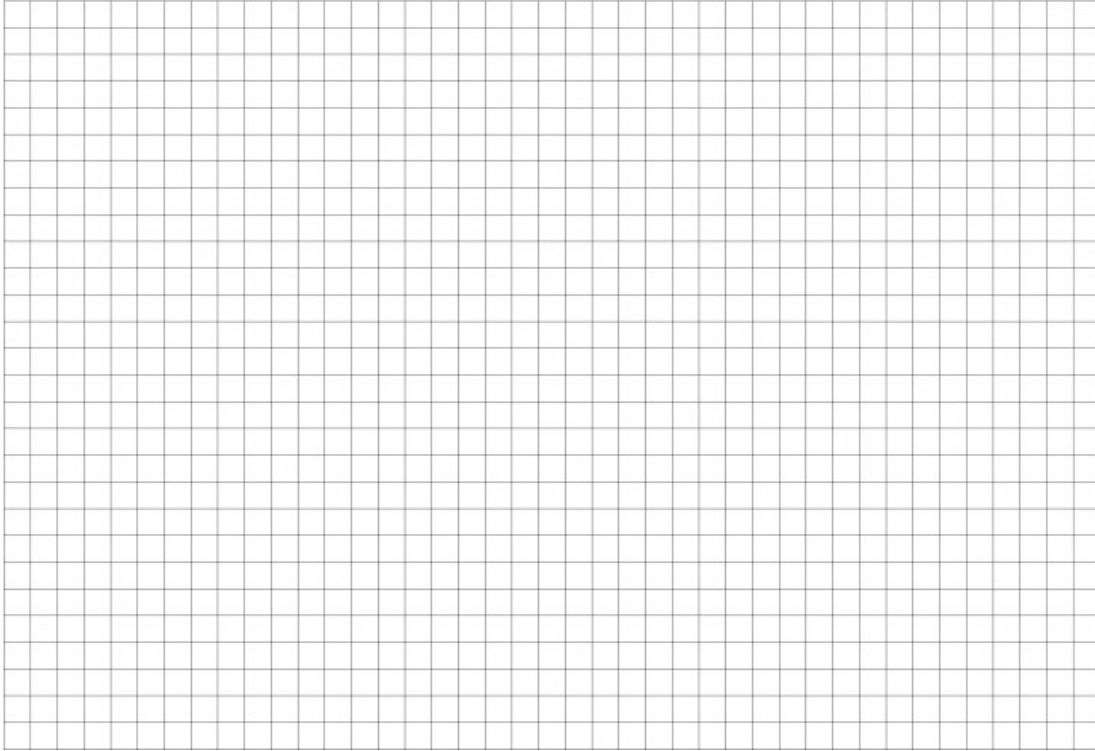


PS at 5 galleons:_____

PS at 10 galleons:_____

Part 3

The Ministry of Magic decided to institute a price ceiling of 10 galleons to allow many more to afford the treat. Use a graph of the demand curve and the supply curve to evaluate the welfare effects this policy had on the pumpkin pastie market.



CS: _____

Δ CS: _____

PS: _____

Δ PS: _____

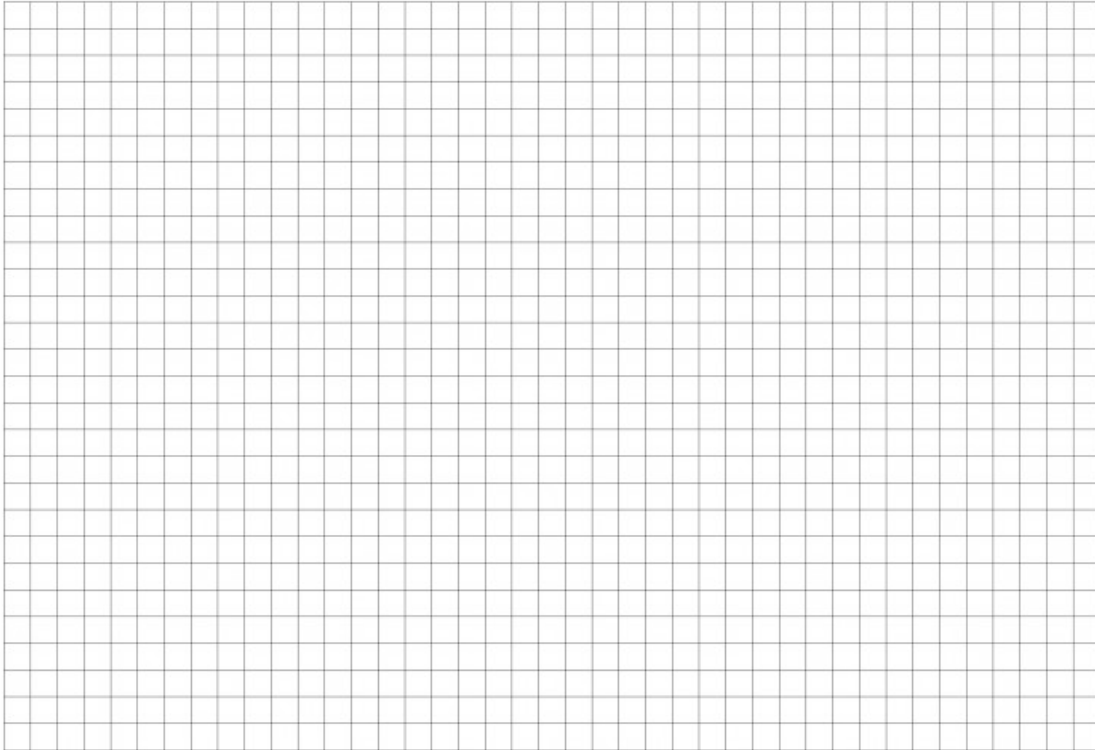
DWL: _____

Shortage/Excess: _____

Part 4

After years of careful epidemiological analysis, a subcommittee of the Ministry tasked with improving the health and wellbeing of the wizarding community published a story in the Daily Profit establishing a link between the consumption of pumpkin pasties and accidental magical spell casting by wizards and witches in public areas, with many cases of innocent muggles being nearly injured. To address these obvious public health concerns, the fiscal arm of the Ministry imposed a 2 galleon

tax on the sale of pumpkin pasties, and reinvested the revenues into researching a magical remedy for this diet-driven ailment.* Use a graph to illustrate the impact this tax had on the market.



CS (post-tax): _____

Eq. Price (pre-tax): _____

Δ CS: _____

Eq. Price (post-tax): _____

PS (post-tax): _____

Eq. Quant (pre-tax): _____

Δ PS: _____

Eq. Quantity (post-tax): _____

DWL: _____

*Note: assume there are no externalities, which we'll cover next week; pumpkin pasties are a private good; the market is competitive.

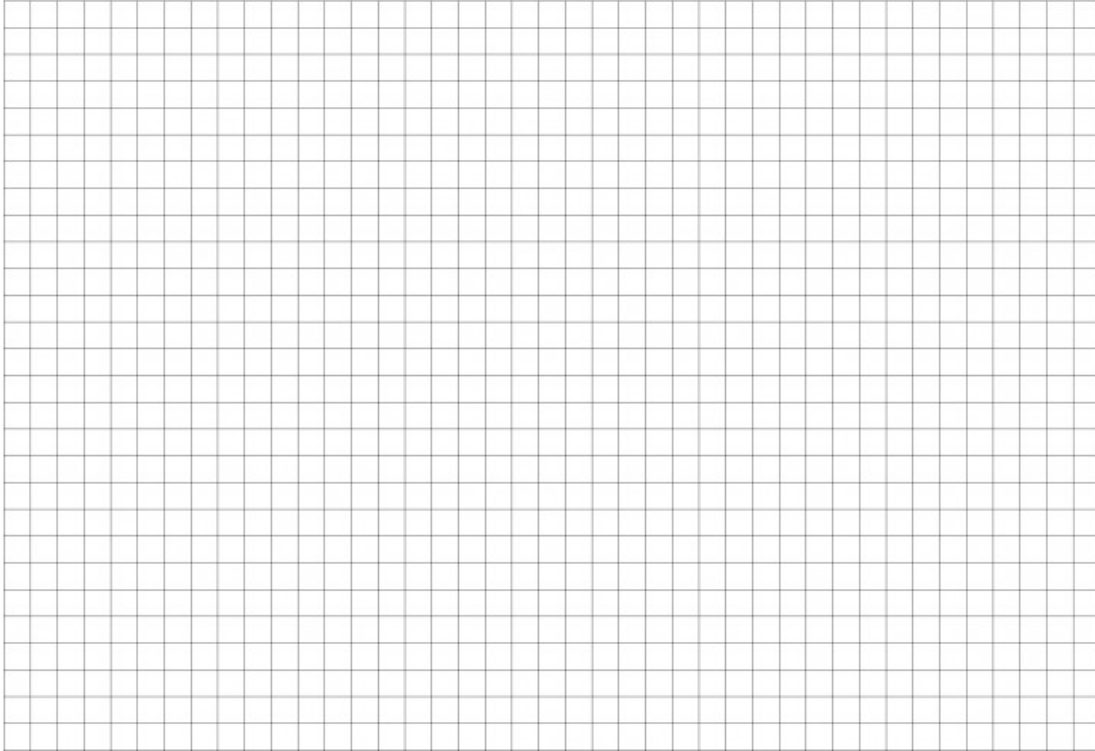
Part 5

Going back to the original supply and demand curves:

$$P = 17 - \frac{1}{6}Q$$

$$P = 2 + \frac{2}{3}Q$$

In recent years, trade barriers were lifted for exports of pumpkin pasties to international magical markets with a world price of 12 galleons. The Ministry of Magic used this opportunity to pad its coffers by imposing an export tariff of 2 galleons. Use a graph to illustrate what happened to the market after the tariff was imposed.



What is the area of the deadweight loss from the tariff? _____